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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Maxillary Sinusitis

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outline

- Anatomy
- Etiologies
- Classifications
- Oroantral fistula
- Management

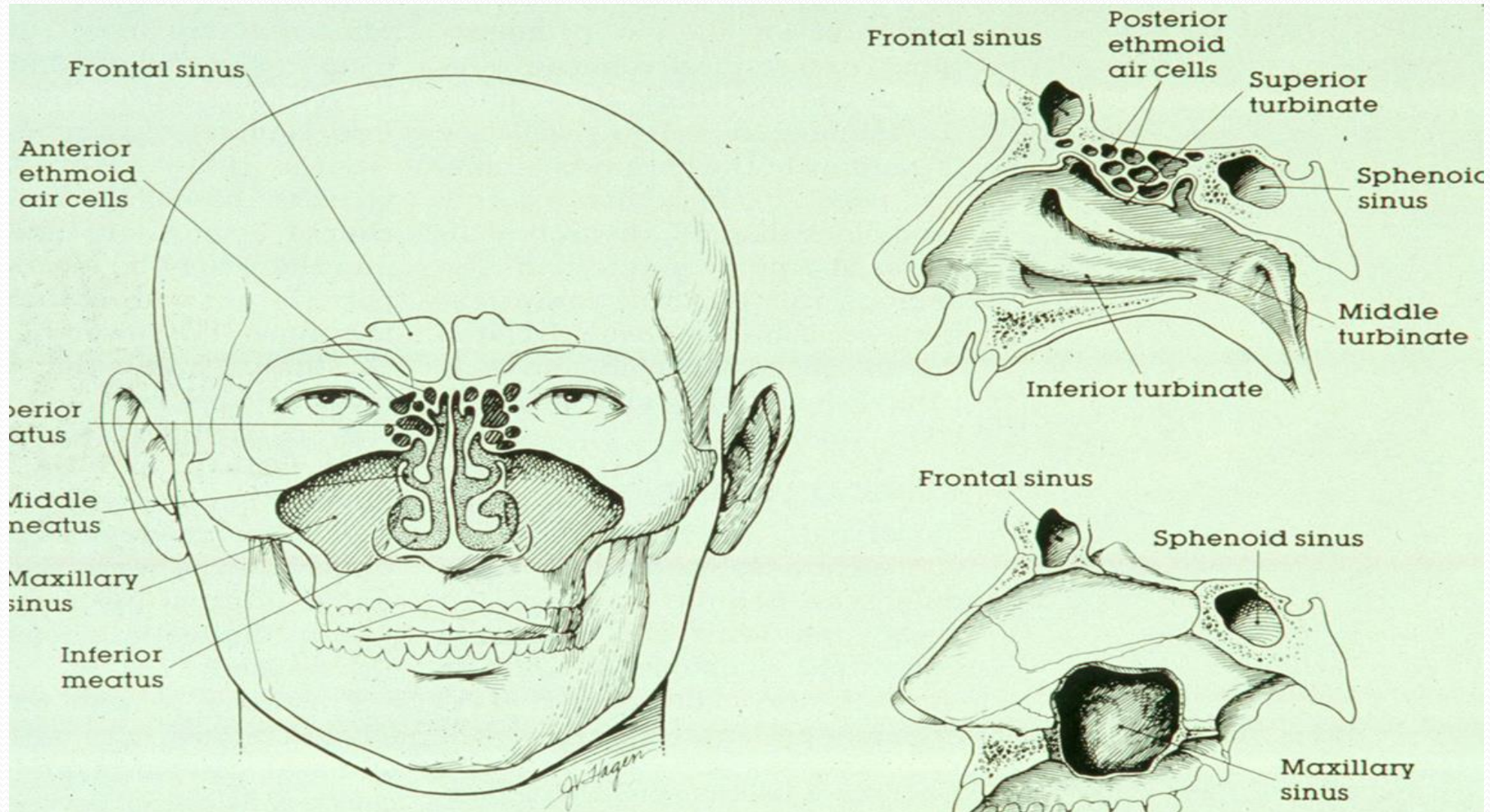
Anatomy of Maxillary Sinus

- The maxillary sinus was first described in 1651, by Nathaniel Highmore.
- Therefore, it is also known as antrum of Highmore.
- Maxillary sinuses are two in number
- They are the largest of the paranasal air sinuses.
- They may be identical or asymmetrical in size and shape.

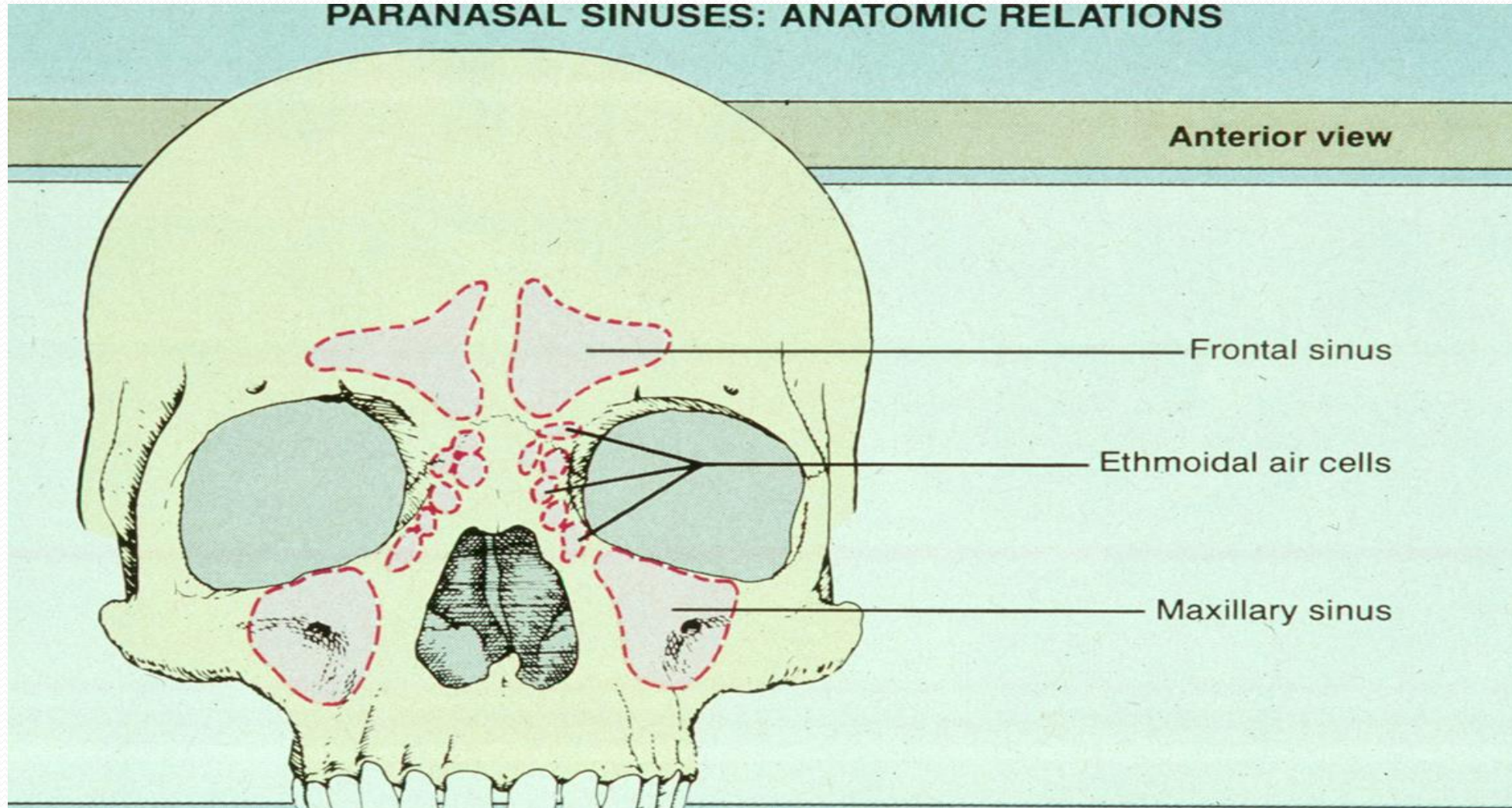
Cont....

- The average dimensions of the sinus are approximately 3.5 (AP) x 3.2 (height) x 2.5 (width) cm (Turner 1902).
- Its volume is 15 to 30 ml.
- The ostium (3 to 6 mm diameter) opens into middle meatus.
- Pyramidal in shape, consisting of a base, an apex and four sides.
- Lining? pseudostratified columnar ciliated epithelium (respiratory type)

Cont.



Cont.



Boundaries

The apex projects laterally into zygomatic process of maxilla; and it may extend into zygomatic bone when the sinus is very large.

The four walls of pyramid are formed by:

- (i) The roof of antrum or the floor of orbit,
- (ii) The anterior
- (iii) Infratemporal surfaces of body of maxilla
- (iv) The alveolar process of maxilla which is the floor of sinus.

EMBRYOLOGY

table 39.1: stages of the growth of maxillary sinus

Time	Growth	Shape
3/12 IU	Outpouching in middle meatus	—
Birth	Tubular: 2 cm × 1 cm × 1 cm	
	3 mm per year × 2 mm × 2 mm	Tubular
9 years	60% of adult size	Ovoid
12 years	Antral floor parallels nasal floor	—
18 years	Adult size	Pyramida

Maxillary sinusitis

- Sinusitis is a clinical diagnosis that is confirmed by physical examination, including nasal endoscopy, and radiographic imaging.
- Although the term *sinusitis* is commonly in use, the process may more accurately be described by the term *rhinosinusitis* because the nasal and sinus mucosal surfaces are contiguous and it would be impossible to have sinusitis without a coexisting rhinitis.

Maxillary sinusitis

- ❖ *Inflammation of most or all of the paranasal air sinuses simultaneously is known as pansinusitis*
- ❖ *Maxillary sinusitis is an inflammation of the maxillary sinuses.*
- ❖ *If infection or an allergic reaction affects the sinus, the antral mucosa may become thickened with pathogenic associated changes, including edema, degeneration, hyperplasia or hypertrophy.*

Classification

- Maxillary sinusitis can be divided into acute and chronic, based on duration of disease, and odontogenic or non-odontogenic based on the etiology.
- Non-odontogenic factors, especially upper respiratory tract infections and allergic reactions, are the most common etiologies.
- with approximately 10% of all maxillary sinusitis cases being of dental origin.

Maxillary sinusitis of odontogenic origin

- Usually manifests unilaterally
- Its pathophysiology, microbiology, and management are different from non odontogenic sinusitis.

Etiology and pathogenesis

- ❖ Periapical abscesses
- ❖ Severe periodontitis
- ❖ Peri-implantitis
- ❖ Post extraction infection

OTHERS

- ❖ Dental or alveolar trauma
- ❖ Odontogenic cysts
- ❖ Maxillary osteomyelitis
- ❖ Iatrogenic or accidental displacement of foreign bodies (e.g. teeth, root tips, dental implants, osseous fragments, bone graft materials, endodontic therapy materials)

Microbiology

Acute non- odontogenic sinusitis

- Predominantly yields aerobic bacteria, in particular *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*, and *Staphylococcus aureus*.

Acute odontogenic sinusitis

- The most frequent bacteria isolated include viridans streptococci and anaerobes such as *Peptostreptococcus*, *Porphyromonas*, *Prevotella*, and *Fusobacterium*.

Acute odontogenic maxillary sinusitis

Common signs and symptoms include

- ⦿ Dull or intense pressure-like pain, erythema, swelling of the cheek and the anterior maxilla.
- ⦿ Pressure or fullness in the vicinity of the maxillary sinus.
- ⦿ Headache, malaise, fever, oral malodor, mucopurulent rhinorrhea, nasal congestion or obstruction, and
- ⦿ Drainage of foul-smelling mucopurulent materials into the nasal cavity and nasopharynx (postnasal drip).

Serious Complications

- Orbital cellulitis
- Cavernous sinus thrombosis
- Meningitis
- Intracranial abscesses

Treatment principles

- Early and aggressive antibiotic therapy and elimination of the source of infection.
- Relief of pain and reduction of the nasal mucosal edema.
- Drainage serves to reduce pain intensity, prevents disease progression, and encourages resolution.
- Elimination of the dental source by tooth extraction, apicectomy, endodontic therapy, removal of any involved foreign body.

Chronic odontogenic maxillary sinusitis

- Local signs and symptoms are generally subtle but malodor, persistent pus discharge from the oroantral fistula .
- Nasal congestion, and discharge, with or without a postnasal drip, are not uncommon.
- Toothache during chewing, increased mobility of teeth, migraine, dull headache, and anterior maxillary tenderness are also possible symptoms.

Treatment

- ✓ Elimination of the infection source.
- ✓ Antibiotic therapy and surgery.
- ✓ If an oroantral fistula is present, frequent irrigation of the sinus cavity via the fistula.
- ✓ Surgical closure

Imaging

- Occipitomenal (OM) 15° (waters) view .
- The presence of pus will produce a horizontal fluid level in this view; provided that there is air above it.
- A CT scan is often considered the gold standard imaging method for the maxillary sinus in developed Countries.

Cont.

- Due to thickened antral mucosa and fluid accumulation, a range of opacification in the affected sinus is often evident.
- Since the disease is usually unilateral, comparison in opacification between opposite sides is often helpful.
- Radiographic Translucency level is generally lower in the affected sinus than in the unaffected site.

PNS VIEW



Trans illumination

- First remove patient's dentures
- Use darkened room
- Shield light source from observer's eyes
- Use Welch Allyn trans illuminator or Mini-Mag Lite
- Shine light over max. sinus & observe light transmission through hard palate
- Report results as opaque, dull, or normal for either side



One way to transilluminate the maxillary sinus is to shine the light at the midpoint of the infraorbital ridge and look for transmission of the light through the hard palate.

Diagnosis

- Evaluation of the patient's signs and symptoms.
- Past medical history, adequate imaging studies, and laboratory examination.
- Rhinoscopy, nasal and sinus endoscopy, and aspiration of sinus contents for cytologic and microbiologic assessments.

Caldwell–Luc procedure

- The Caldwell–Luc procedure involves the complete removal of the antral lining and creation of a new opening for more dependent drainage into the nose by the transoral approach.

Indication's

1. Open procedure for removal of root fragments, teeth or foreign body or an antrolith (stone) from the maxillary sinus.
2. To treat chronic maxillary sinusitis with hyperplastic lining and polypoid degeneration of the mucosa.
3. Removal of cysts or benign growths from the maxillary sinus.

Cont.

4. Management of hematoma in the maxillary sinus and to control post traumatic hemorrhage in the sinus.
5. Zygomaticomaxillary complex fractures involving floor of the orbit and anterior wall of the maxillary sinus.
6. Removal of impacted canine or impacted third molar.
7. Along with closure of chronic oroantral fistula, associated with chronic maxillary sinusitis.

Caldwell-Luc Surgical Procedure

- I. The surgical procedure can be performed under LA with sedation or under GA—which is the preferred method.
 - II. A semilunar incision is planned in the buccal vestibule from canine to second molar area, just above the gingival attachment.
 - III. A mucoperiosteal flap is elevated with the help of periosteal elevator till the infraorbital ridge.
- Care is taken to prevent injury to infraorbital nerve.

Cont.

IV. An opening or window is created in the anterior wall of the maxillary sinus with the help of chisels, gouges or dental drills.

V. The opening is enlarged (about index finger size)carefully in all directions with Rongeur forceps, to permit the inspection of the sinus cavity.

VI. The opening or window created should be well away from the apices of the roots of the maxillary teeth.

Cont.

VII. A thorough irrigation of the maxillary sinus is carried out with copious saline wash.

VIII. Inspection of the maxillary sinus is done and removal of root, stone, etc. can be done at this stage.

IX. The thickened, infected lining of the maxillary sinus can be elevated with Howarth's periosteal elevator and removed and sent for histopathological examination.

Cont.

X. If there is profuse bleeding, then the sinus can be packed with ribbon gauze soaked in adrenaline 1:1000 for 1 or 2 minutes.

XI. The antral cavity again is irrigated and can be packed with iodoform ribbon gauze. The end of the same can be removed through the nasal antrostomy or through the small incision in the buccal vestibule.

XII. The incision is closed with 3-0 silk.

Maxillary sinus in Dentoalveolar Surgery and Trauma

Oro-antral fistula

Invasion of the maxillary sinus and establishment of a direct communication with the oral cavity is referred to as an oro-antral fistula.

Fistula

- ✓ Is a biological tract that connect an anatomical cavity with the external surfaces or another anatomical cavity, (*unlike sinus tract*).
- ✓ It is always lined with a stratified squamous epithelium and the patency of the tract is preserved until epithelial cells scraped off.

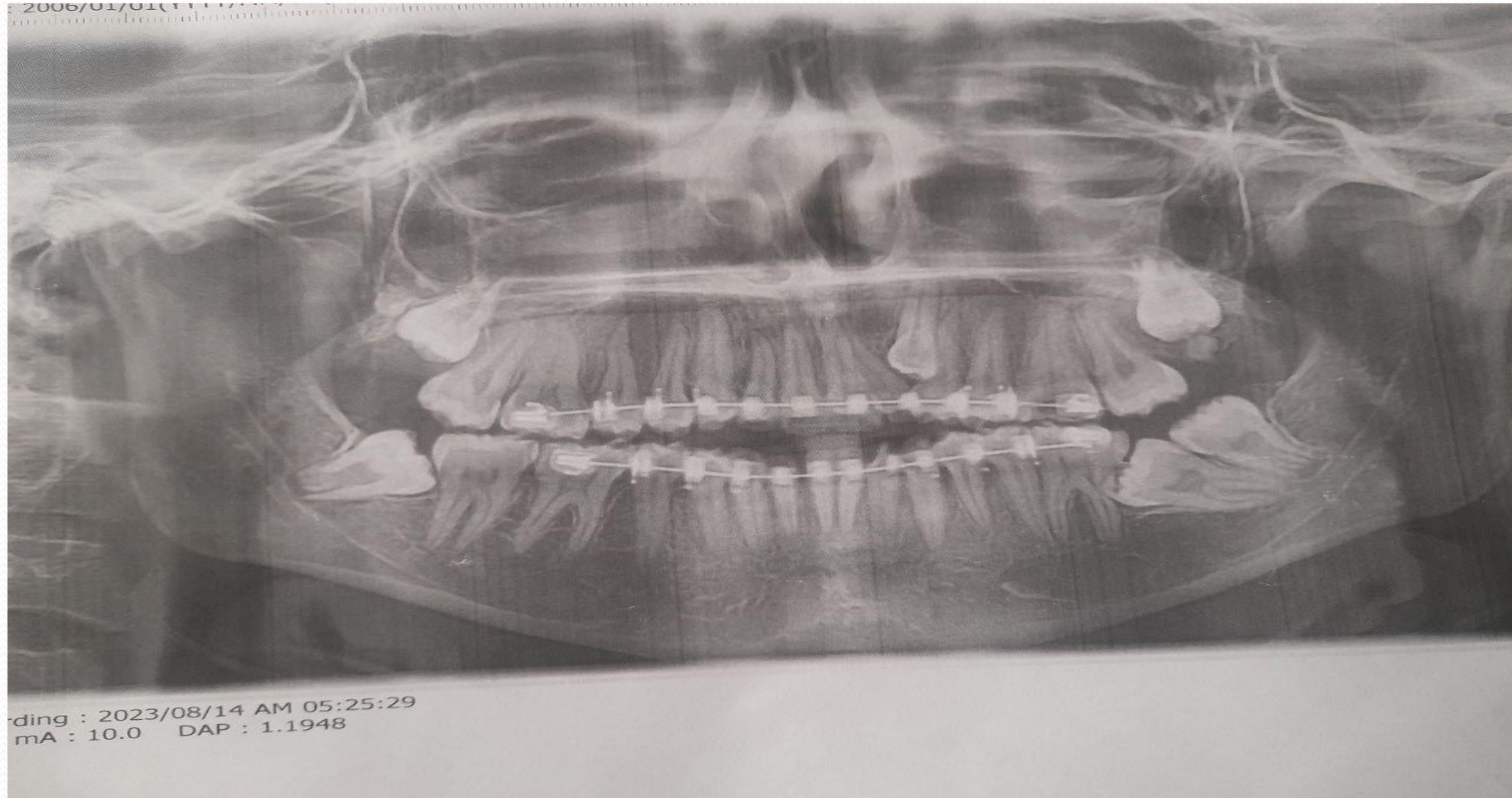
Factors influencing creation of oro-antral fistula

- Teeth size and configuration of the roots.
- Hypercementosis and bulbous roots.
- Density of alveolar bone and thickness of sinus floor
- Size of the sinus.
- Relation of sinus to the root of upper teeth.
- Rough extraction and misguided manipulation.
- Apical pathosis and attached granulomas.
- Periodontal diseases which may erode sinus floor.
- Presence of cysts and neoplasm.
- Invasive surgery e.g. cleft and dental implants placement.

Relation of sinus to the root of upper teeth



Relation of sinus to the root of upper teeth



Signs and symptoms of newly created oro-antral communication

- Antral floor attached to roots apices of extracted tooth or teeth.
- Fracture of the alveolar process or the tuberosity.
- Evidence of air stream passing from nostril.
- Bubbling of blood from the socket or nostril.
- Change in speech tone and resonance.
- Radiographical evidence of sinus involvement.

Confirmation of existence of oro-antral communication

- Instruct patient to occlude the nostrils and blow gently “nose-blowing’ test”.
- If nose-blowing’ test is negative, don’t explore the opening with suction tip and/or probes.
- Always check radiograph for the continuity of sinus floor and presence of entrapped foreign body.

Displacement of tooth or root into the maxillary sinus lining or the sinus cavity proper

- It is basically a mishap incident results from a neglected act by the operator while applying wrong force.
- Occurs rarely but the 2nd, 3rd molar and 2nd premolar are the most at risk of dislodgment.
- May occur during forceful mouth opening of unconscious patient when using mouth gag of periodontally involved teeth.
- May occur with severe maxillofacial injuries.
- In association with poor surgical technique.

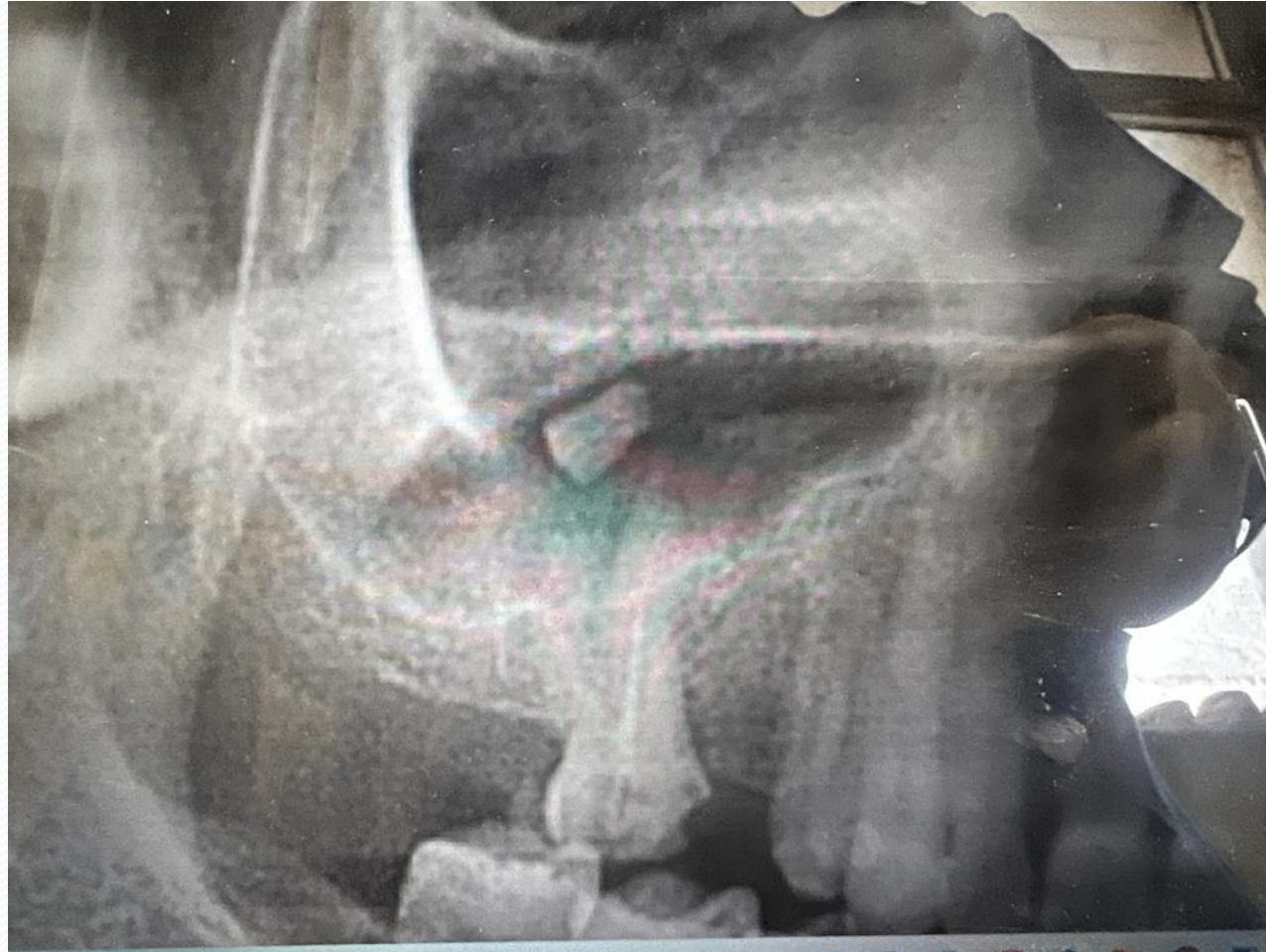
Immediate management/ investigations

- Confirm the existence of oro-antral fistula and the presence of tooth or root in sinus using periapical/occlusal, panoramic and occipito- mental radiographs.
- Locate the precise position of the foreign body within the sinus lining or in the sinus cavity proper “head-shaking test”.

OPG SHOWING DISPLACED LEFT MAXILLARY SECOND MOLAR



OPG SHOWING DISPLACED ROOT OF MAXILLARY SECOND MOLAR

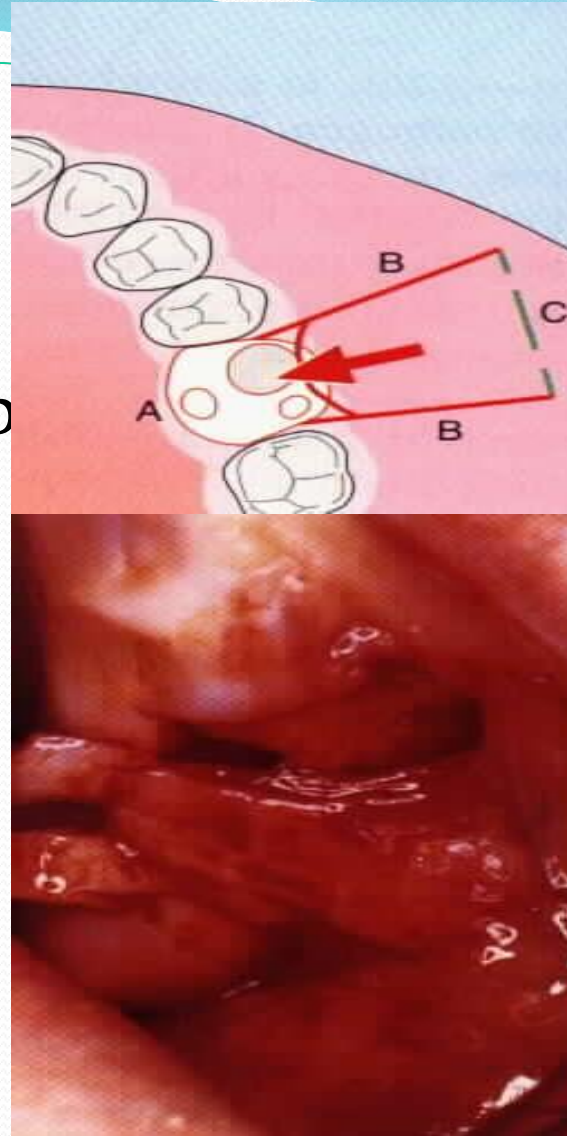


Immediate management/ foreign body retrieval

- Reflect mucoperiosteal flap.
- Reduce alveolar bone height.
- Retrieve the tooth or the root by permitting their movement away from the sinus.
- If root or tooth dislodged into the sinus proper, consider Caldwell-luc approach.
- Undermine the flap and replace across the bony defect.

Immediate management/ closure of the defect

- Relieve the tension of the flap by severing the periostium.
- Advance the flap across the defect and beyond.
- Anchor the corner of the flap and approximate the edges using horizontal mattress sutures.



Postoperative care/ Home care

- Acrylic base plate may be prescribed to add additional support to the area.
- Patient should avoid forceful nasal blowing, if forced to do so, no occluding of nares.
- Oral hygiene must be kept optimum.

Postoperative care/ medications

- Antibiotic
e.g. Penicillin or penicillin derivatives
- Analgesic
- Decongestant
e.g. Ephedrine or otrivin nasal drops
3 drops/ 3times daily / 7 days
- Steam inhalation
e.g. menthol and benzoin

Chronic oro-antral fistula/ persistent oro-antral communication

It might be a complication of:

- Unrecognized (overlooked) fistula.
- Untreated fistula.
- Failure of surgically repaired fistula

Signs and symptoms of chronic fistula

- Reflux of food and drinks.
- Loss of denture stability.
- Intermittent episode of pain and local tenderness.
- Foul-tasting discharge.
- Signs and symptoms of chronic sinusitis.



Primarily management of chronic OAF

- ▶ It is aimed to eliminate any sinus infection:
- Excision of any mucosal polyp or purulent granulation to promote drainage.
- Regular irrigation with warm water or saline.
- Single course of antibiotics and nasal inhalation and decongestant.
- Acrylic base plate.
- Surgical closure



Surgical management/ Principles and requirements

- Flap should have good blood supply.
- Flap tissue must be handled gently.
- Flap should lie in its new position without tension.
- Good haemostasis must be achieved

Buccal Advancement Flap

- This technique is used primarily for closure of a minor alveolar communication .
- It is not the preferred method for large communications or recurrent fistulae.
- Secondly, because the flap is thin, wound dehiscence might occur after the operation.
- Thirdly, this technique may result in a loss of vestibular depth.

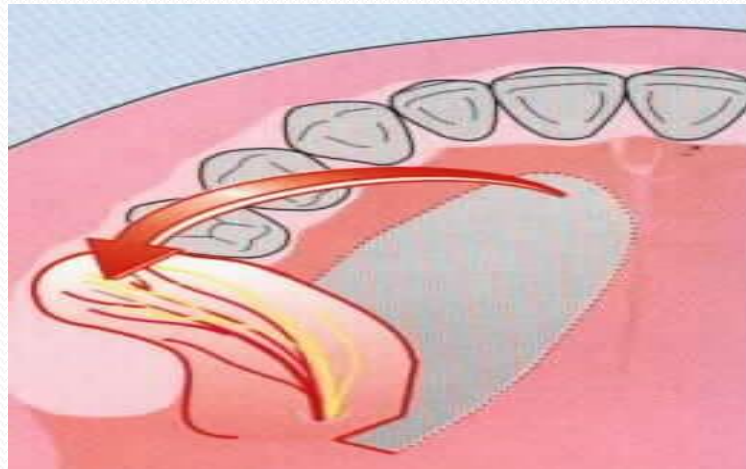
Surgical management/ types of repair

➡ Buccal advancement flap



Palatal flap technique

- The palatal flap harbors the greater palatine vessels that provide a rich blood supply to the flap.
- This technique involves mobilization and rotation of large mucosal flaps to cover the osseous defect with soft tissues, the margins of which are sutured over.



Advantages

- A. Provision of more tissue attachment without tension, which promotes healing and resists the tendency of the flap to return to its original position.
- B. Use of a flap that is firmer and more resistant to trauma and infection
- C. A broad spectrum of indication, including cases with large bone defects; and
- D. Preservation of the maxillary vestibular sulcus depth.

Disadvantages

- A. Denudation of the palatal surface.
- B. Possible greater postoperative pain.
- C. It is a relatively complicated technique that require extensive clinical experience of the surgeon.
- D. The later appearance of roughness and deepening of the donor area, resulting from secondary epithialization.
- E. Possible flap necrosis if the artery is injured.
- F. Interference with wearing of a partial denture that covers the hard palate.

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Thank You